Recall the inverse of e^x is $\ln(x)$. That is if

$$e^y = x, \quad \ln(x) = y.$$

Take $y = \ln(x)$, from implicit derivatives

$$e^y = x$$

$$y' e^y = 1$$

$$y' = \frac{1}{e^y}$$

$$y' = \frac{1}{x}$$

So,

$$\boxed{\begin{aligned} &\frac{d}{dx}(\ln(x)) = \frac{1}{x}, \quad x > 0 \\ &\frac{d}{dx} \ln|x| = \frac{1}{x} \quad \text{all } x \in \mathbb{R} \end{aligned}}$$

From the chain rule

$$\frac{d}{dx}[\ln(g(x))] = \frac{g'(x)}{g(x)}$$

Ex: $\frac{d}{dx}(\ln(\sin x)) = \frac{\cos x}{\sin x} = \cot x$

Integration:

We can now evaluate more integrals with the fact that

$$\int \frac{dx}{x} = \ln|x| + C$$

Ex:

$$\int \tan(x)\,dx = \int \frac{\sin(x)}{\cos(x)}\,dx \quad , \quad \text{let } u = \cos x, \; du = -\sin x\,dx$$

$$= -\int \frac{du}{u} = -\ln|u| + C$$

$$= -\ln|\cos x| + C$$

$$= \ln|(\cos x)^{-1}| + C$$

$$= \ln\left|\frac{1}{\cos x}\right| + C$$

$$\boxed{= \ln|\sec x| + C}$$